Chang Gung University

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智慧運算學院
College of Intelligent Computing

Self-Supervised Masked Autoencoders for High-Accuracy Left Ventricle Segmentation in Echocardiography

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ACML 2025
MEDICAL AI WORKSHOP

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PhD Program, Division of Artificial Intelligence, Graduate Institute of Clinical Medical Sciences
Jointly offered by College of Intelligent Computing & College of Medicine

Agenda

01 Clinical Motivation

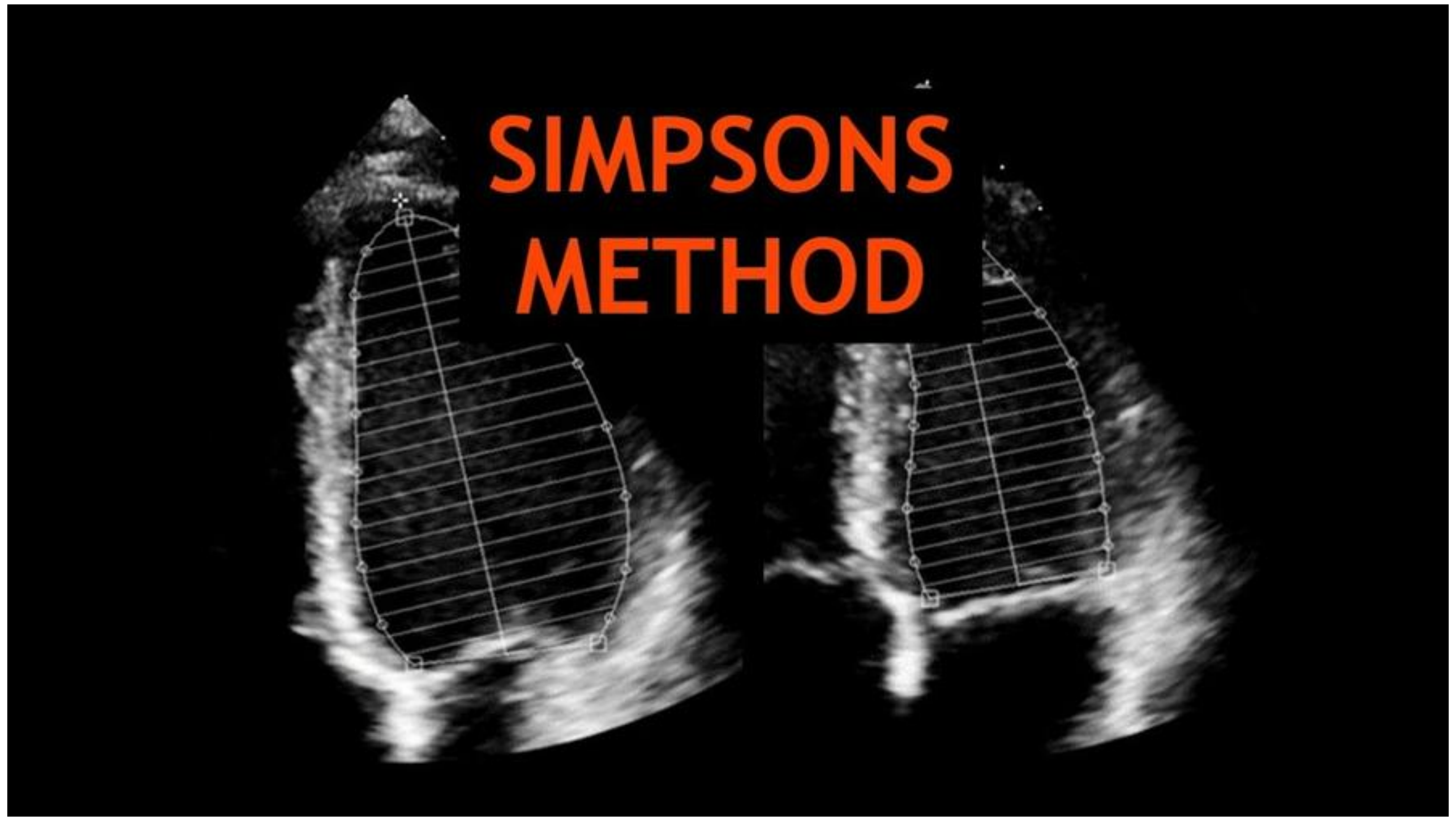
02 AI Methodology

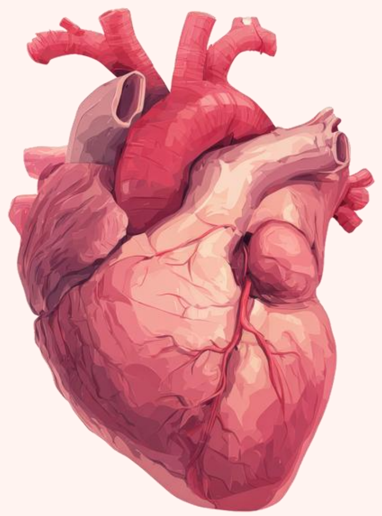
03 Experimental Results

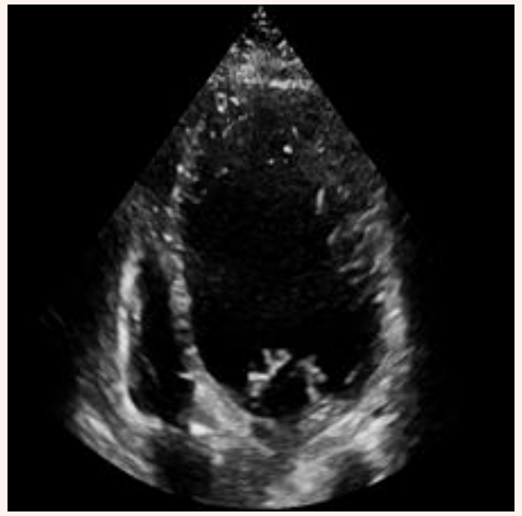
04 Conclusion & Future Work

Clinical Motivation: The Challenge of LV Segmentation

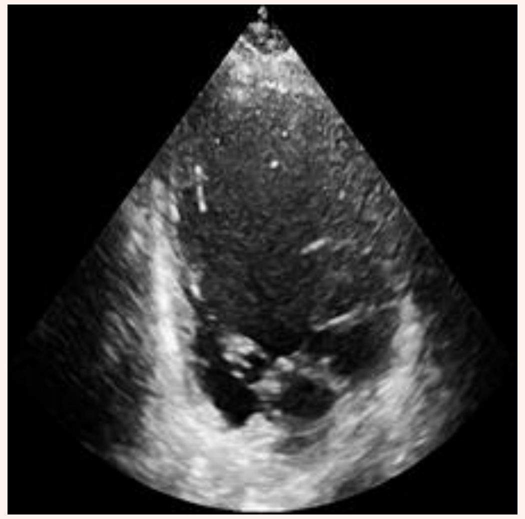
Guideline standard for 2D EF







A4C



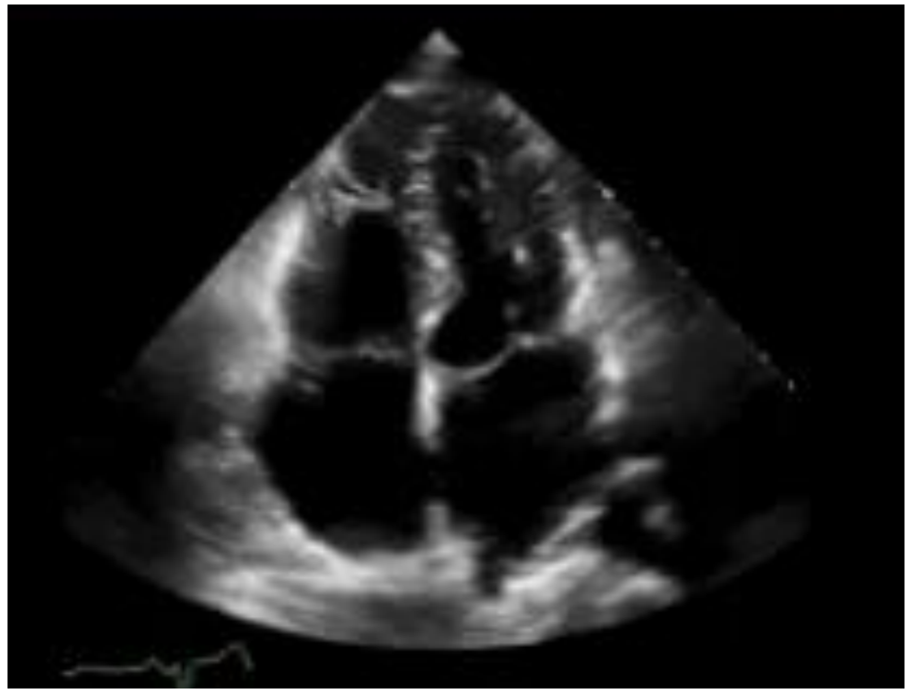
A2C

The Challenge: Manual tracing is time-consuming and labor-intensive.

Baseline Model: EchoNet-Dynamic[1]

Stanford AIMI

Center for Artificial Intelligence in Medicine and Imaging



Scale

10,030 labeled videos from Stanford University.

Quality

Rigorous human expert annotations (tracings & calculations)

Architecture

DeepLabv3 (CNN-based) with temporal consistency.

Resolution

Operates at 112 × 112 (Low Resolution)

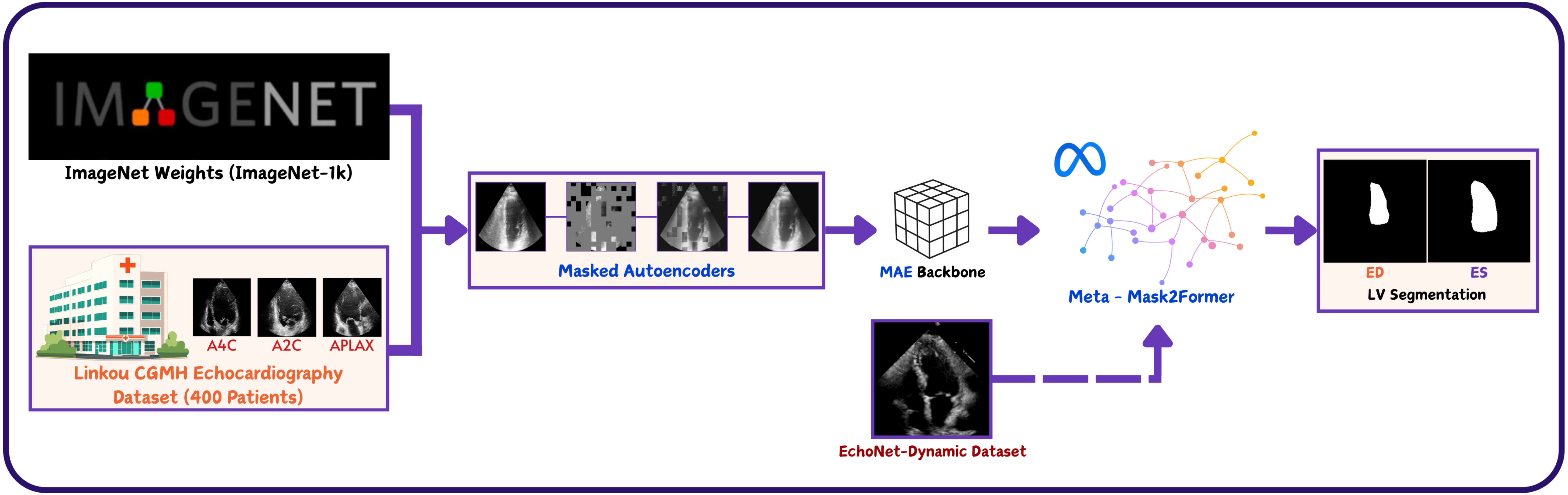
Task

Automated Beat-to-Beat EF assessment.

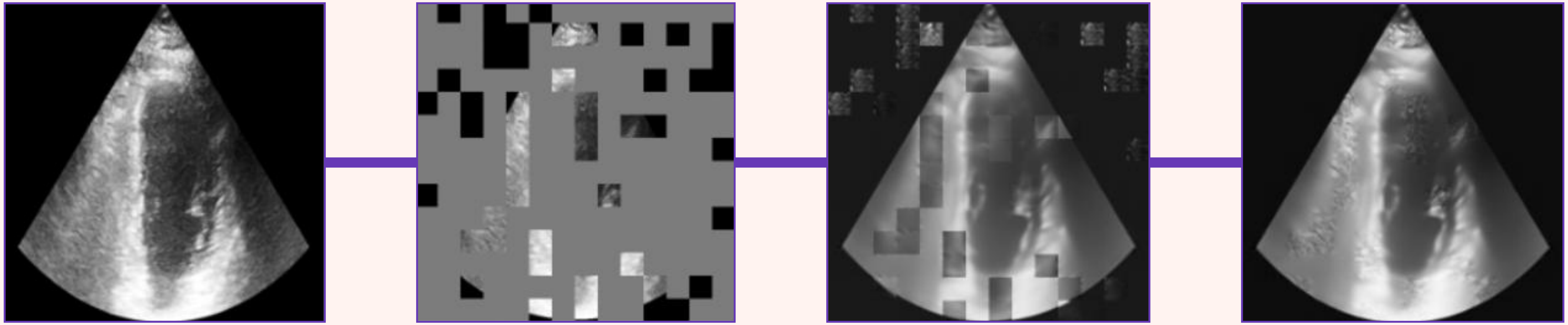
nature

[1] Ouyang, D., He, B., Ghorbani, A. et al. Video-based AI for beat-to-beat assessment of cardiac function. Nature 580, 252–256 (2020). https://doi.org/10.1038/s41586-020-2145-8

AI Methodology: Proposed Framework



AI Methodology: Self-Supervised Pre-training (MAE[2])



Original

Masked Input (75%)

Reconstructed Patches

Full Reconstruction

High Masking Ratio (75%) forces the model to learn global context and robust features.

[2] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, Ross Girshick; Masked Autoencoders Are Scalable Vision Learners. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 16000-16009

AI Methodology: Segmentation Head(Mask2Former)[3]

Meta - Mask2Former

Dice Loss

Maximizes global shape overlap (IoU)

Cross-Entropy Loss

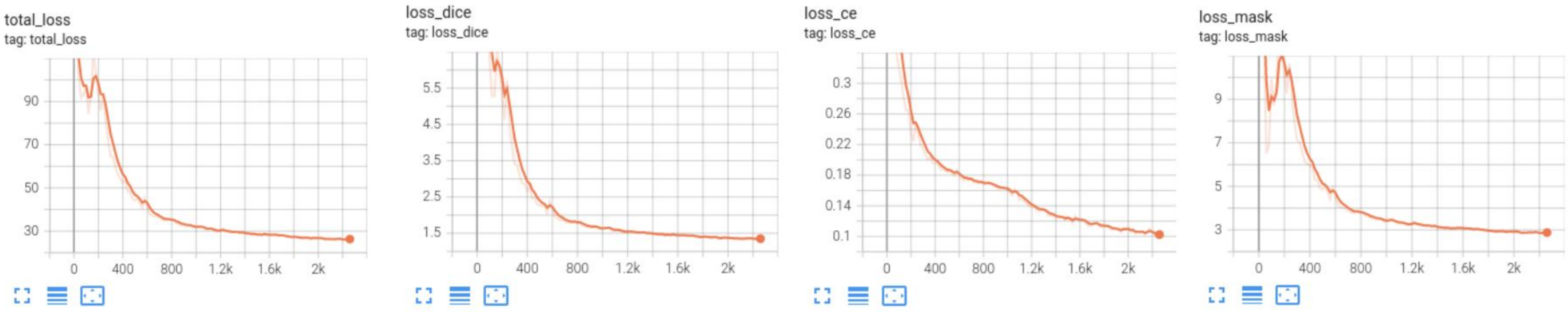
Predicts the correct class
(LV vs. Background)

Mask Loss

Refines pixel-level binary boundaries

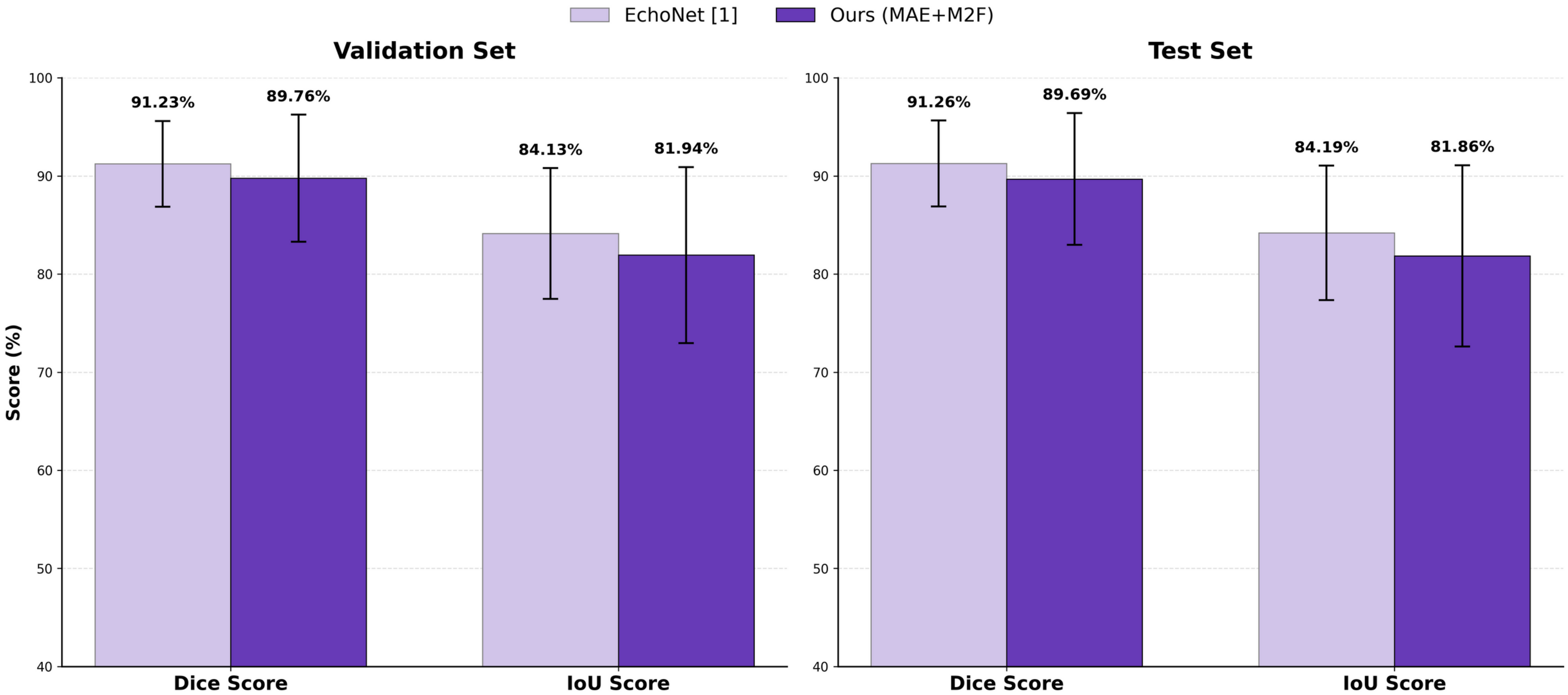
[3] Bowen Cheng, Ishan Misra, Alexander G. Schwing, Alexander Kirillov, Rohit Girdhar; Masked-Attention Mask Transformer for Universal Image Segmentation. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 1290-1299

AI Methodology: Training Analysis



MAE pre-training accelerates convergence and enhances training stability.

Experimental Results



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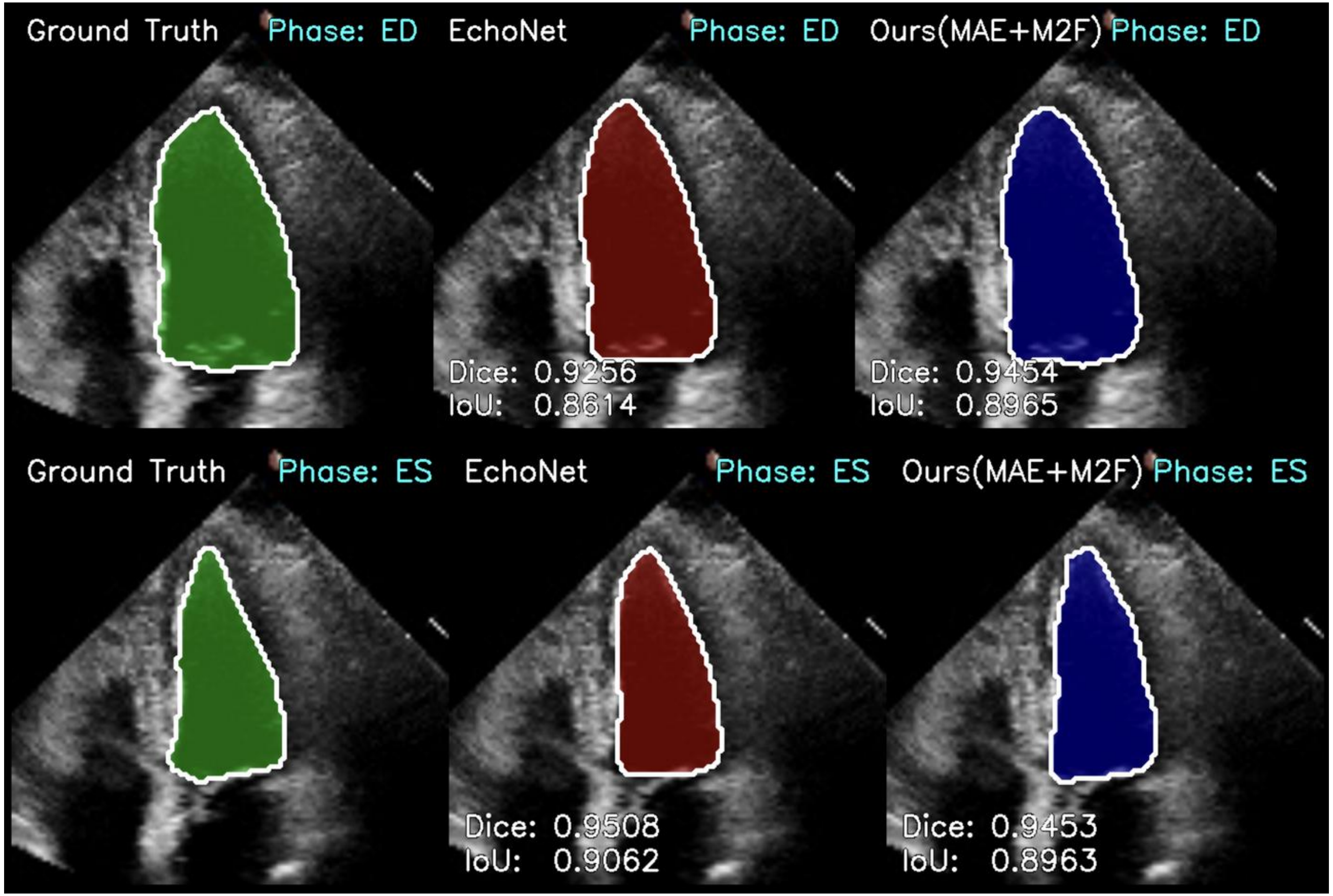
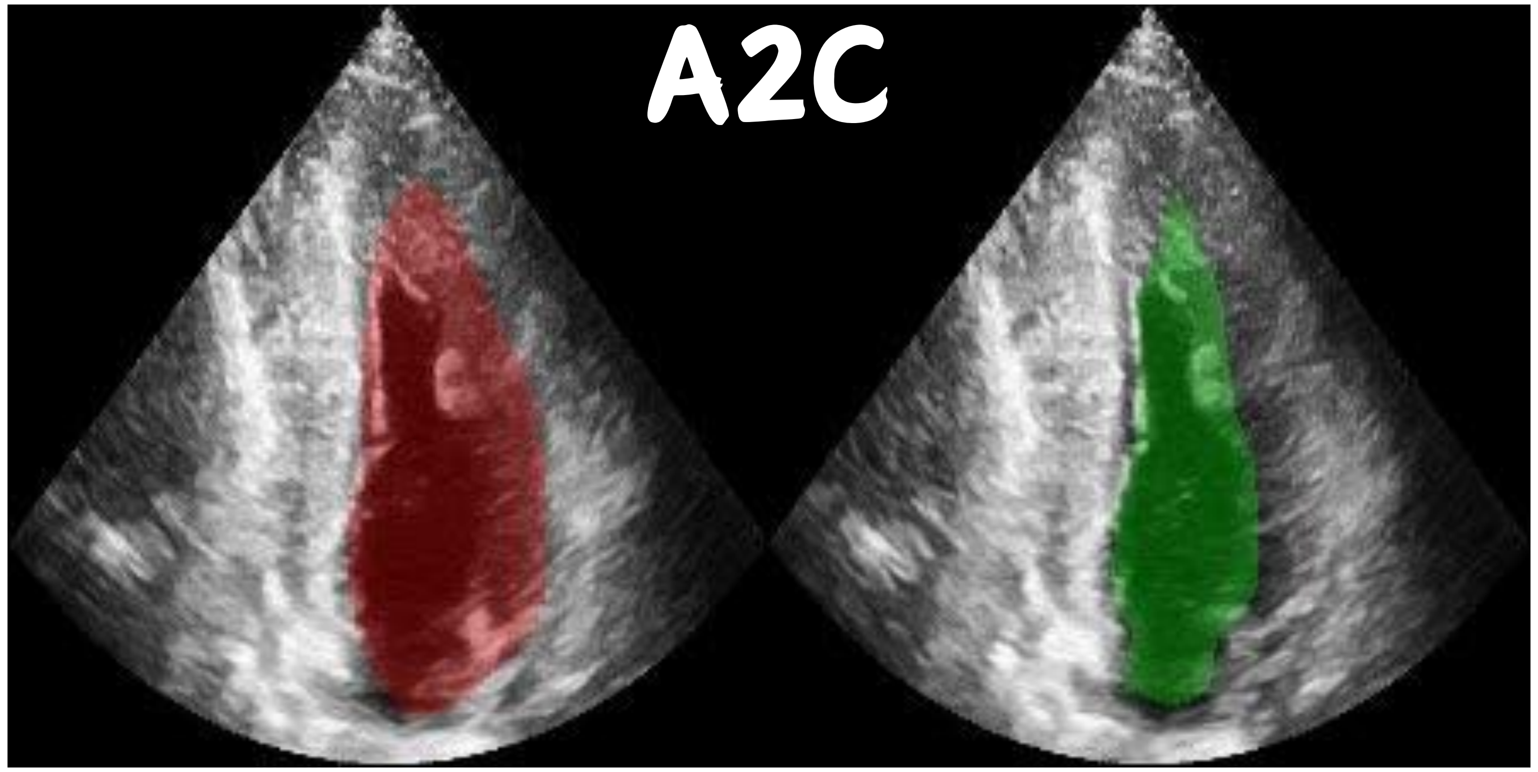
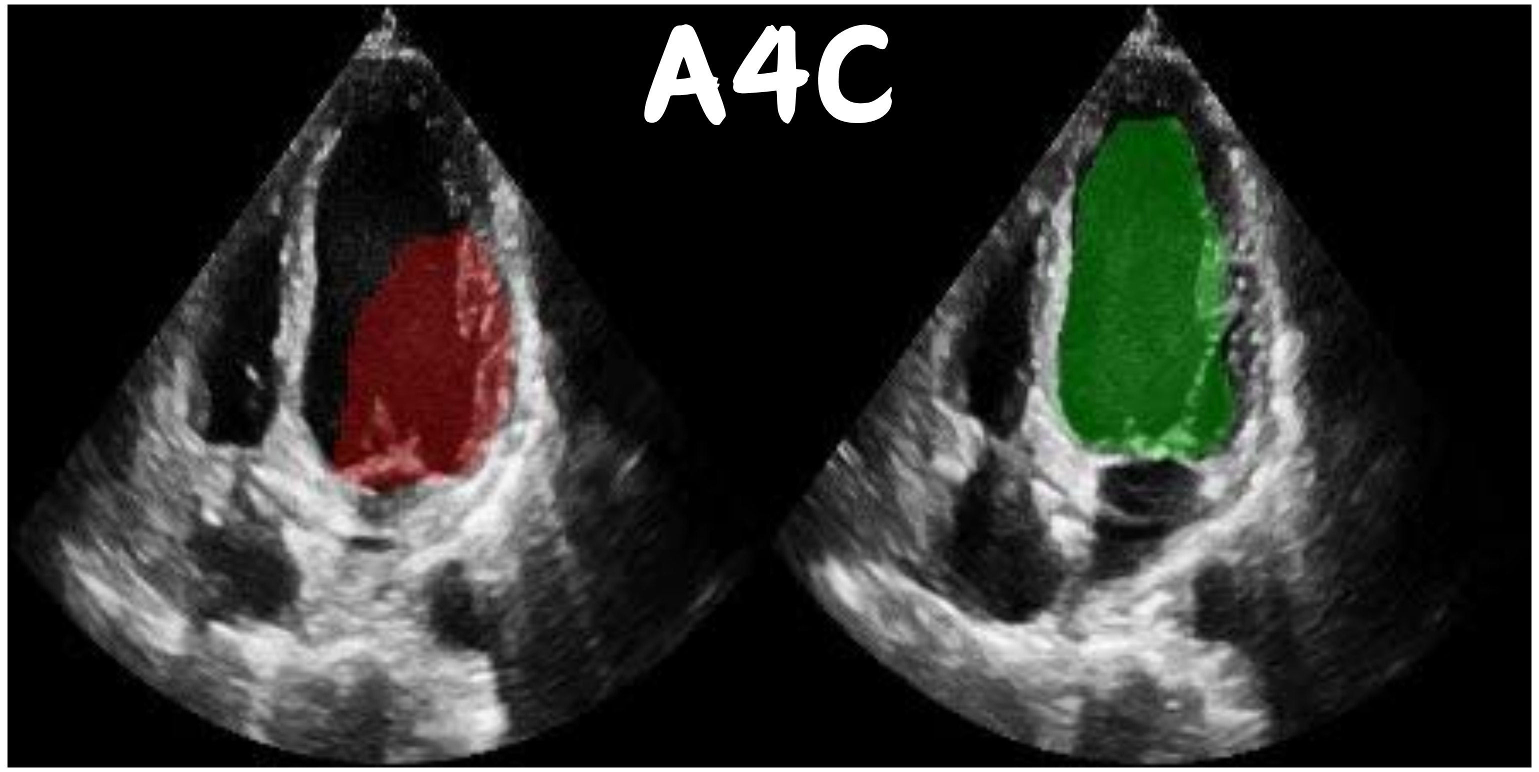
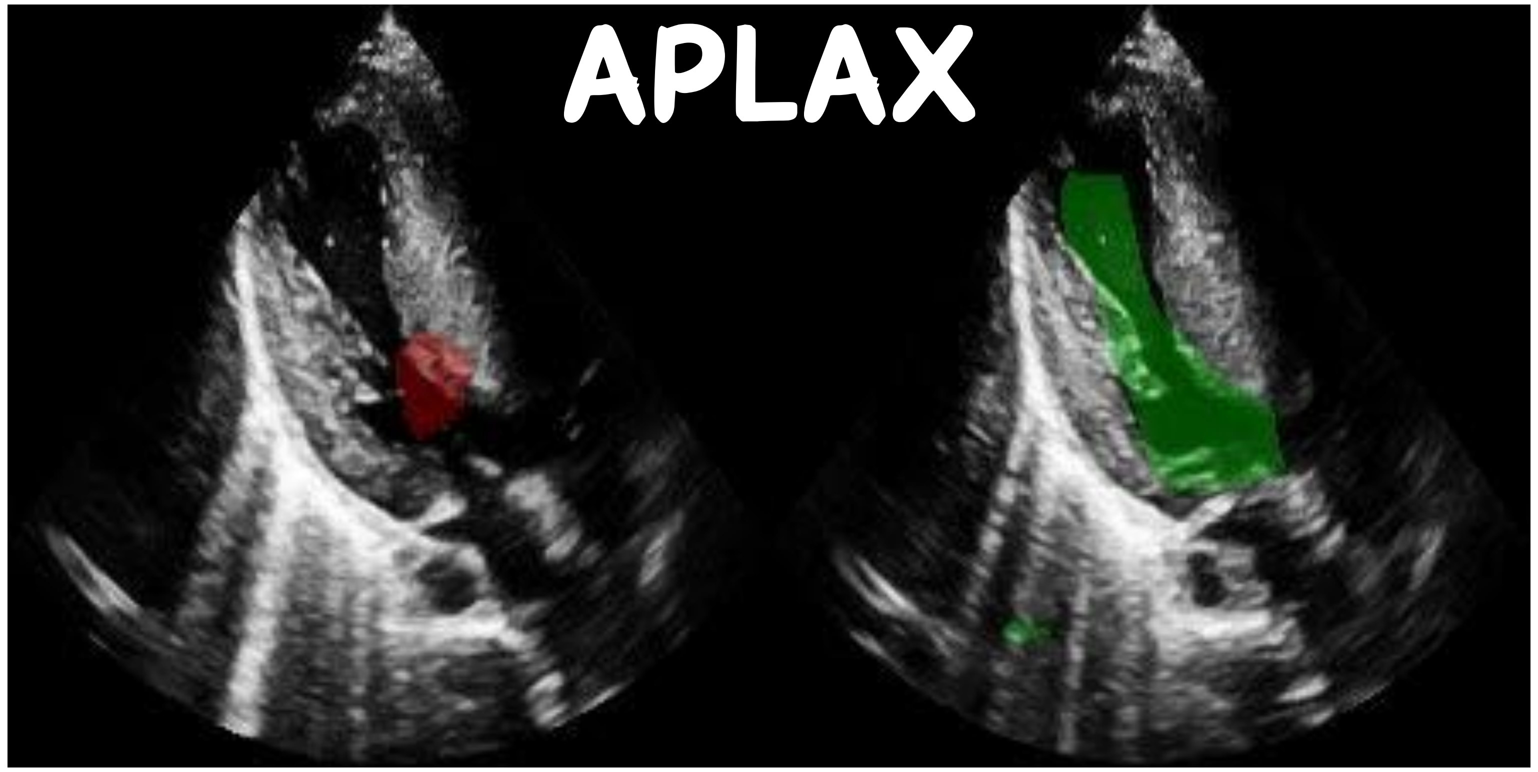
Experimental Results (EchoNet-Dynamic Dataset)



Experimental Results (LinKou CGMH Dataset)



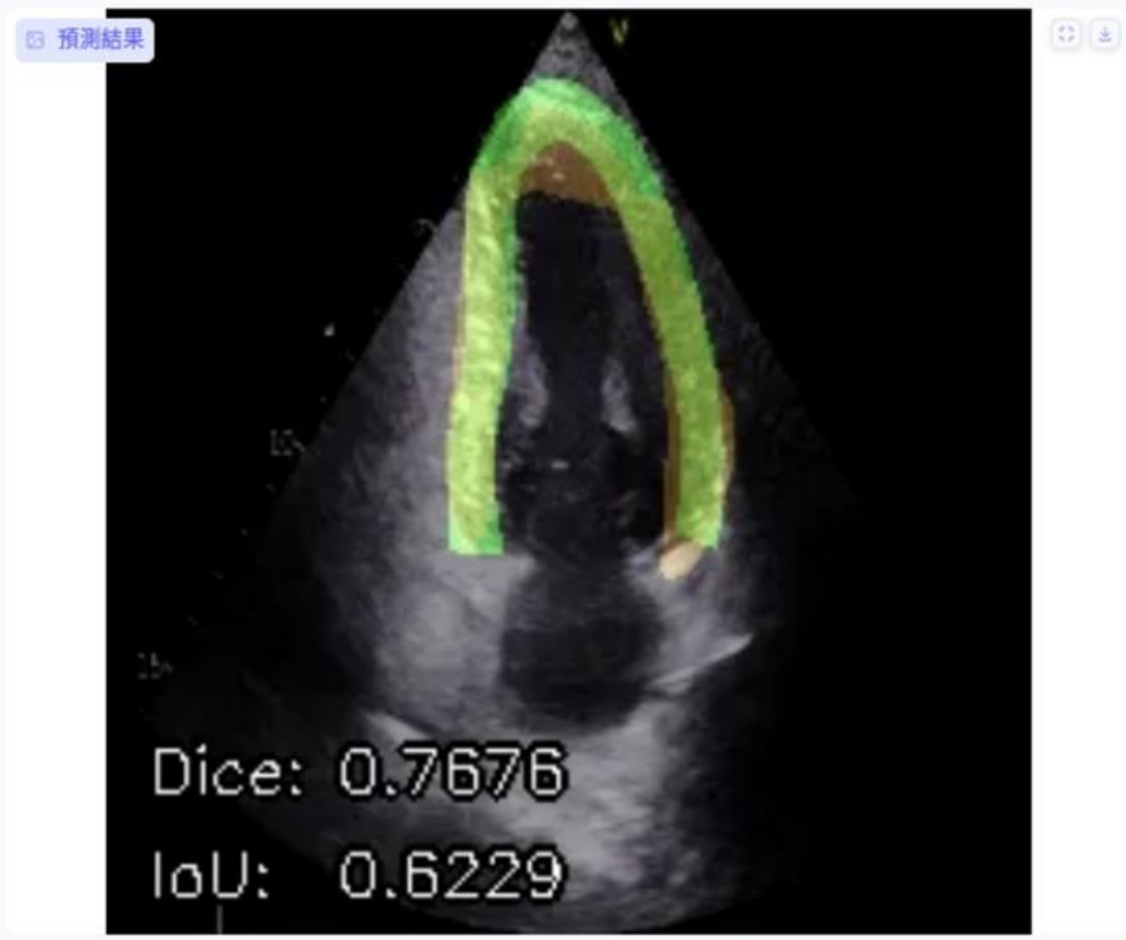




Red: EchoNet

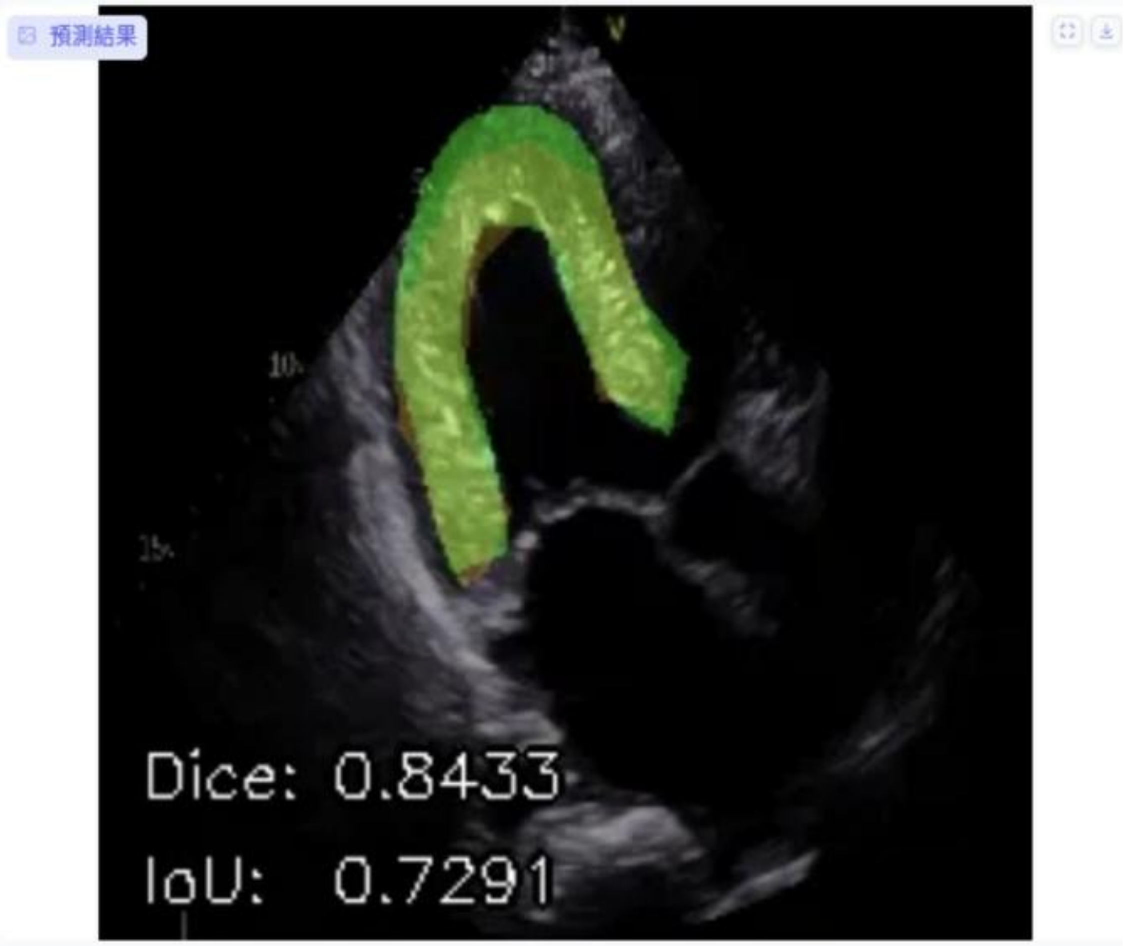
Green: Ours Model

Future Work



A2C

A4C



APLAX

Target: Global Longitudinal Strain (GLS)(Standard 3-Apical Views)

Conclusion

Robustness: Self-supervised pre-training (MAE) significantly enhances feature extraction robustness against ultrasound noise.

Efficiency: The MAE pre-trained backbone facilitates rapid convergence and improves training stability.

Competitive Strategy: Transfer Learning combined with Continued Pre-training achieves clinical equivalence to the SOTA EchoNet baseline.

Clinical Impact: This framework provides a highly accurate, automated tool for cardiac function assessment, facilitating precise clinical decision-making.

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Thank You

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